

Industrial Internship Report on " Student Management System Using Java Full Stack Development"

Prepared by

DIVAKAR M

Executive Summary

This report presents the details of the **Industrial Internship in Java Full Stack Development** provided by **upskill Campus and The IoT Academy** in collaboration with **UniConverge Technologies Pvt Ltd (UCT)**.

The internship provided practical exposure to Java programming, JDBC, database connectivity, and the fundamentals of full-stack application development. During the training, I worked through Java Fundamentals, JDBC concepts, and Full Stack Development topics along with related assessments.

As part of the internship, I developed the concept of a **Student Management System Using Java Full Stack Development**. The project focuses on organizing and managing student information through a structured application, demonstrating the application of Java and database connectivity concepts learned during the internship.

This internship provided an opportunity to understand how technical concepts can be applied to practical application development. It also helped improve my programming knowledge, problem-solving ability, database understanding, and overall awareness of the Java Full Stack development process.

The internship was a valuable learning experience that helped connect academic knowledge with practical software-development concepts.

TABLE OF CONTENTS

1	Preface	4
2	Introduction	5
2.1	About UniConverge Technologies Pvt Ltd	5
2.2	About upskill Campus	9
2.3	Objective	10
2.4	Reference	11
2.5	Glossary	11
3	Problem Statement	13
4	Existing and Proposed solution	14
5	Proposed Design/ Model	15
5.1	High Level Diagram (if applicable)	15
5.2	Low Level Diagram (if applicable)	17
5.3	Interfaces (if applicable)	18
6	Performance Test	19
6.1	Test Plan/ Test Cases	19
6.2	Test Procedure	20
6.3	Performance Outcome	20
7	My learnings	21
8	Future work scope	23

1 Preface

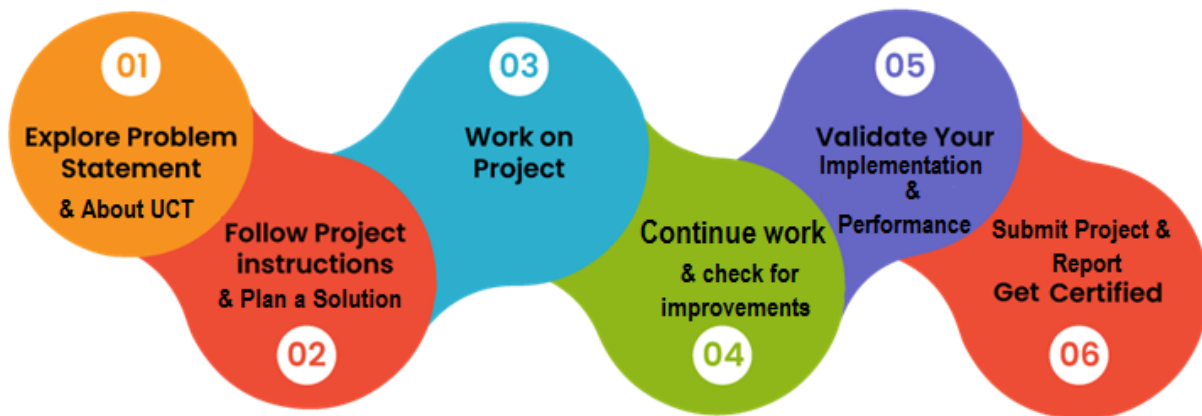
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



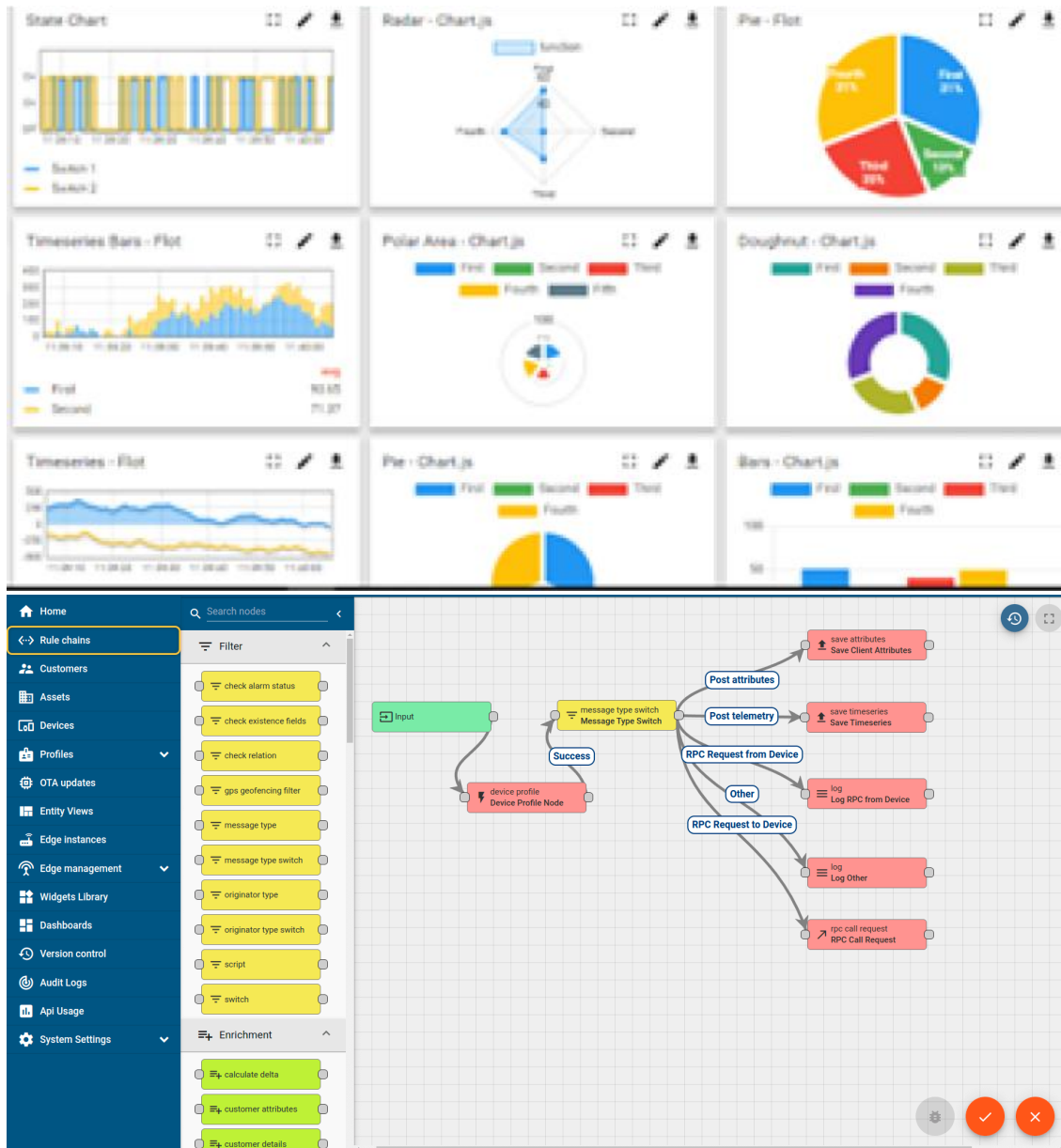
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



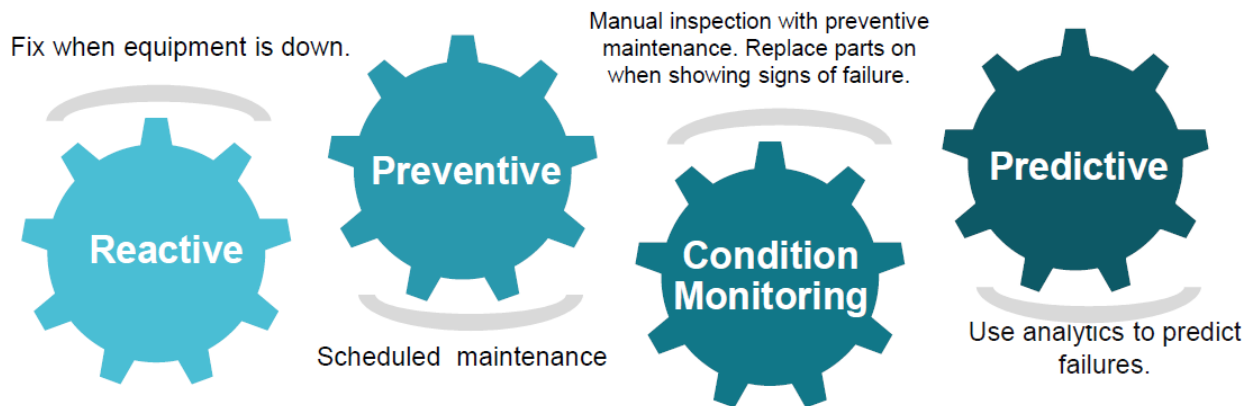


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

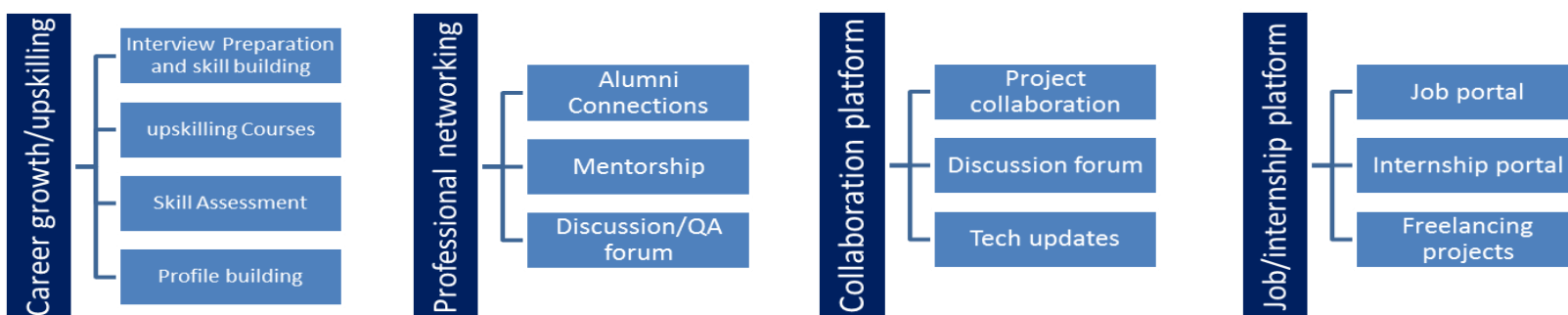


2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.





2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] **UniConverge Technologies Pvt. Ltd. (UCT)** – Internship and technology-related information provided as part of the training program.
- [2] **upskill Campus** – Training materials and learning resources provided during the internship program.
- [3] **The IoT Academy** – Learning resources and technical guidance provided as part of the internship.

2.6 Glossary

Terms	Acronym
Java	A high-level, object-oriented programming language used to develop software applications.
Java Full Stack Development	Development of both the front-end and back-end components of a web application.
JDBC	Java Database Connectivity, an API used to connect Java applications with databases.

Database	An organized collection of data that can be stored, accessed, and managed electronically.
MySQL	A relational database management system used to store and manage application data.

3 Problem Statement

Managing student information manually can be time-consuming and may make it difficult to maintain, update, and retrieve student records efficiently. A structured application is required to organize student details and provide basic operations for managing those records.

The **Student Management System Using Java Full Stack Development** is proposed to address this requirement. The system is designed to provide a simple application through which student information can be added, viewed, updated, and deleted in a systematic manner.

The project applies the concepts learned during the internship, particularly **Java programming, JDBC, database connectivity, and Full Stack Development**. The internship template identifies the Problem Statement as a specific section of the project report, followed by the Existing and Proposed Solution sections.

4 Existing and Proposed solution

Existing Solution

In a traditional student record management process, student information may be maintained using **manual records or basic spreadsheet-based methods**. These approaches can be useful for maintaining small amounts of information, but managing a larger number of records can become difficult.

Proposed Solution

The proposed **Student Management System Using Java Full Stack Development** provides a structured application for managing student information. The system applies the Java programming and JDBC/database concepts covered during the internship.

4.1 Code submission (Github link):
<https://github.com/Divakar12455/upskillCampus->

4.2 Report submission (Github link) :
<https://github.com/Divakar12455/upskillCampus->

5 Proposed Design/ Model

The proposed **Student Management System Using Java Full Stack Development** is designed as a structured application for managing student information. The design follows a layered approach in which the user interface communicates with the application logic, which in turn interacts with the database.

The internship template specifies that this section should explain the design flow and include the **High-Level Diagram, Low-Level Diagram, Interfaces, block diagrams, data flow, protocols, flow charts, and related design details** where applicable

5.1 High Level Diagram

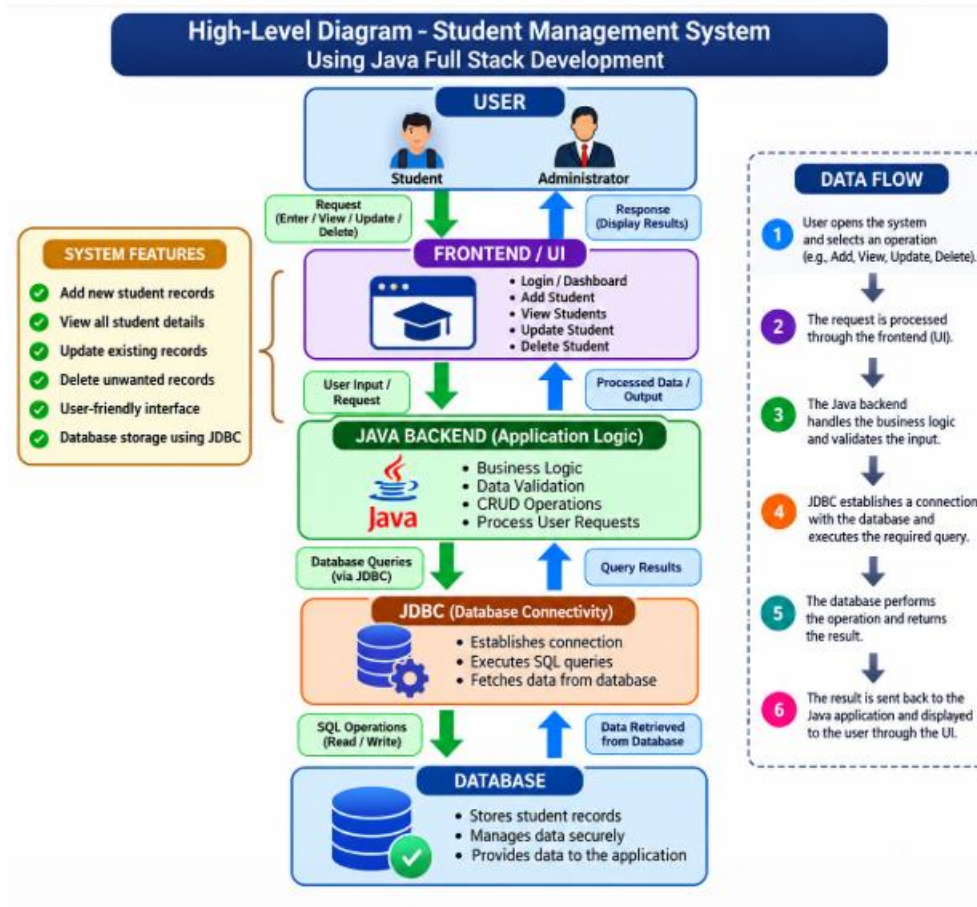
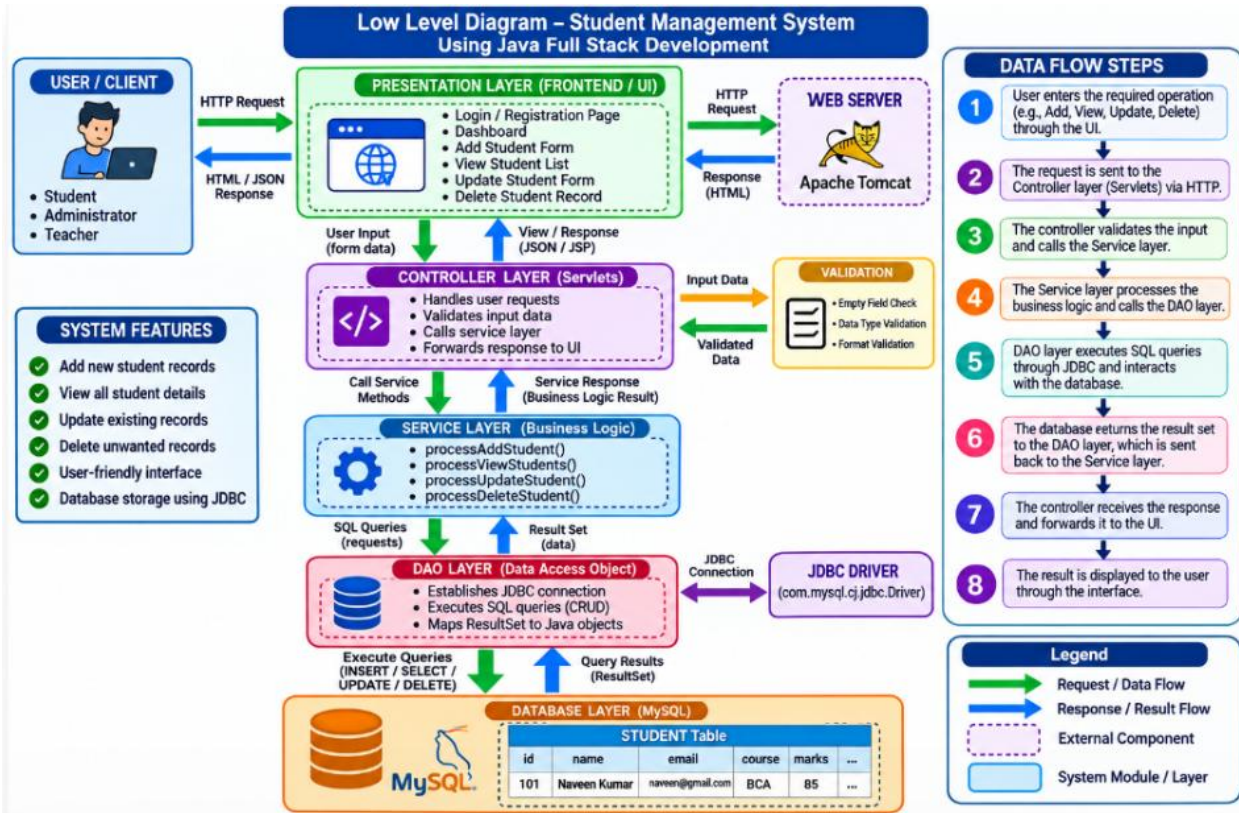


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram



5.3 Interfaces

1.3

Interfaces

Student Management System
Using Java Full Stack Development

The following section shows the main interfaces of the Student Management System. These interfaces allow the user to perform all basic operations such as login, view, add, update and delete student records through a simple and user-friendly interface.



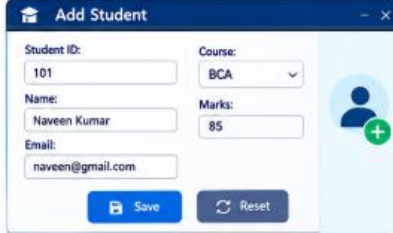
1. Login Interface

Allows the user to enter their credentials (username and password) to access the system.



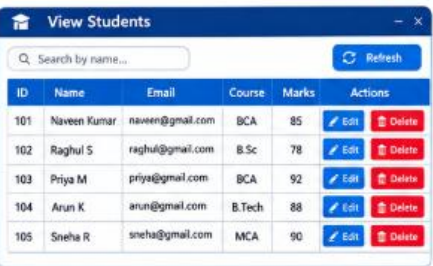
2. Dashboard Interface

Displays the main menu and quick access to all available operations in the system.



3. Add Student Interface

Provides a form to enter new student details and store them in the database.



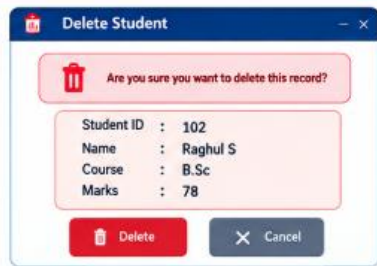
4. View Students Interface

Displays the list of all student records stored in the database with options to edit or delete.



5. Update Student Interface

Allows the user to modify existing student details.



6. Delete Student Interface

Confirms and deletes the selected student record from the database.

6 Performance Test

The **Performance Test** evaluates the Student Management System in terms of functionality, response time, stability, resource usage, and database operations. The internship template specifies that this section should describe performance constraints, the methods used to handle them, test cases, test procedures, and performance outcomes.

6.1 Test Plan/ Test Cases

1. Test Plan

Parameter	Details
Objective	To verify the performance, stability and response time of the Student Management System.
Environment	Windows 10, Java 17, MySQL 8.0, Apache Tomcat 10
Tools Used	JMeter (load testing), Browser (Chrome), MySQL Monitor
Test Types	Functional, Load, Stress, and Response Time Test
Load Scenario	Simulated 100 concurrent users performing CRUD operations
Evaluation Metrics	Response time, Throughput, Error rate, CPU & Memory usage

2. Test Cases

ID	Test Case	Input Data	Expected Result	Status
TC-01	Login with valid credentials	Username: admin Password: admin123	User should be logged in successfully	✓ Passed
TC-02	Add new student	Name: Naveen Kumar Course: BCA, Marks: 85	Record added to database and displayed in the list	✓ Passed
TC-03	View all students	No input	All student records displayed within 1 second	✓ Passed
TC-04	Update student details	Student ID: 101 New Marks: 90	Record updated successfully	✓ Passed
TC-05	Delete student record	Student ID: 102	Record deleted from database	✓ Passed

6.2 Test Procedure

- Start the Student Management System.
- Enter the required user information.
- Perform the selected student-management operation.
- Verify the application response.
- Check whether the corresponding database operation is completed.
- Repeat the tests with different student records.
- Record the observed results and identify any errors.

6.3 Performance Outcome

The performance evaluation focuses on whether the application performs its main operations correctly and consistently. The test results should be recorded based on the **actual execution environment and measurements**. No unsupported numerical performance values are assumed in this report.

Observed Outcome

- Login functionality was checked.
- Student records could be added through the application.
- Stored student information could be retrieved.
- Existing student details could be updated.
- Student records could be deleted.
- Database connectivity was verified through JDBC.
- The system was evaluated for correct responses during the tested operations.

7 My learnings

During the **Industrial Internship in Java Full Stack Development**, I gained practical knowledge of Java programming, JDBC, database connectivity, and the fundamentals of full-stack application development. The internship helped me understand how programming concepts can be applied to develop a practical application.

Key Learnings

1. Java Programming

- Strengthened my understanding of Java fundamentals.
- Learned to apply Java concepts while developing application logic.
- Improved my programming and problem-solving skills.

2. JDBC and Database Connectivity

- Learned the purpose and working of JDBC.
- Understood how a Java application communicates with a database.
- Learned the basic process of executing database operations from a Java application.

3. Full Stack Development

- Gained an understanding of frontend and backend components.
- Learned how different application layers communicate with each other.
- Understood the basic flow from user interface to backend and database.

4. Student Management System

- Learned how to organize student information within an application.
- Applied CRUD concepts such as adding, viewing, updating, and deleting records.
- Understood the importance of structured data management.

5. Testing and Debugging

- Learned to test application functionality systematically.
- Improved my ability to identify and resolve errors during development.
- Understood the importance of validating application operations.

6. Problem-Solving Skills

- Developed a more systematic approach to solving programming problems.
- Learned to break a larger application requirement into smaller modules.

7. Professional Skills

- Improved my technical learning ability and confidence.
- Gained practical exposure to project-based software development.
- Understood the importance of continuous learning and applying technical concepts to practical problems.

8 Future work scope

The **Student Management System Using Java Full Stack Development** can be further improved and extended with additional features in future versions. The internship template includes **Future Work Scope** as the final section of the project report.

9 Possible Future Enhancements

1. User Authentication

- Add secure login and registration functionality.
- Provide different access levels for administrators and other users.

2. Advanced Student Management

- Add student attendance, academic performance, and course details.
- Provide options for searching and filtering student records.

3. Improved User Interface

- Develop a more responsive and user-friendly interface.
- Improve navigation and presentation of student information.

4. Database Enhancement

- Expand the database structure to manage additional student-related information.
- Improve data validation and consistency.

5. Report Generation

- Provide downloadable student reports.
- Generate academic and attendance reports based on stored information.

6. Search and Filtering

- Add advanced search options based on student ID, name, course, or other available fields.

7. Deployment

- Deploy the application on a suitable server or cloud environment so that it can be accessed through a web browser.

8. Security Improvements

- Implement stronger authentication, authorization, input validation, and secure data handling.

9.1.1 Conclusion

The project provides a foundation that can be expanded into a more comprehensive student-management application. Future development can focus on improving functionality, usability, security, database capabilities, and deployment while applying additional Java Full Stack Development concepts.